

Problem-Solution Fit

Date	04 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

The framework below ensures the proposed solution accurately addresses the root causes, behaviors, and limitations of the target audience for the Credit Card Approval Prediction system.

1. CUSTOMER SEGMENT(S) [CS] - Prospective credit card applicants (salaried & self-employed) - Bank credit analysts reviewing incoming applications	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited digital literacy among some applicants - Incomplete financial documentation at submission - Analysts' limited daily manual-review capacity	5. AVAILABLE SOLUTIONS PROS & CONS [AS] - Manual credit review — human judgement, but slow & inconsistent - Generic credit-scoring software — automated, but not tuned to the bank's own data, costly licensing
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Applications delayed 3-5 days awaiting manual review (daily) - High, inconsistent rejection rate from ad-hoc risk judgement (recurring)	9. PROBLEM ROOT / CAUSE [RC] - No automated first-pass risk score - Multi-class payment-status data isn't simplified into a clear risk signal for reviewers	7. BEHAVIOR + ITS INTENSITY [BE] - Applicants repeatedly check status online / call support (high intensity) - Analysts manually cross-reference income, credit history & past-due records per file (frequent)
3. TRIGGERS TO ACT [TR] - Rising application volume outpacing manual review capacity - Growing customer complaints about approval delays	10. YOUR SOLUTION [SL] An ML-based approval prediction system — Logistic Regression, Random Forest, XGBoost, and Decision Tree models are trained and compared, and the best performer is deployed via a Flask web app + IBM Watson Machine Learning, returning instant approve/reject predictions.	8. CHANNELS OF BEHAVIOR [CH] ONLINE Bank's web & mobile application portal OFFLINE In-branch assistance from a credit analyst
4. EMOTIONS BEFORE / AFTER [EM] Before: Anxious, uncertain, frustrated by the wait After: Relieved and confident with an instant, transparent decision		